EXPERIENCES.FUN

Explore Globe

Agent-Ready Product & Technical Specification

A globe-first geographic discovery layer for Experiences.fun. The globe is not decoration. It is the interface for answering one question: Where is activity happening?

Product relationship

Feed = WHO is doing something • Globe = WHERE it is happening • Experience = HOW to participate

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1. Product Objective

Add a 3D interactive globe to the Explore page that becomes the geographic discovery layer of Experiences.fun. Users move from a global view into countries, cities, local communities, nodes and individual experiences without being overwhelmed by thousands of markers.

• Discover experiences happening around the world.

• See where community activity is concentrated.

• Discover verified nodes such as schools, libraries, community centres, studios, NGOs, universities and businesses.

• Understand local participation, XP and contribution activity.

• Move from geography directly into an actionable experience or location page.

• Create a visual foundation for future impact, reputation and activity layers.

Core principle: Do not build a pretty Earth with some dots on it. Build a geographic activity interface.

2. Explore Page UX

Hero structure

```
EXPLORE
Discover what's happening around the world

[ 3D GLOBE ]
London • Mogadishu • Lagos • Nairobi

NEAR YOU
Experiences happening nearby

TRENDING LOCATIONS
Communities with momentum

UPCOMING EXPERIENCES
```

The globe should be the visual hero of Explore. Supporting cards remain useful, but the globe must not become a background image behind a conventional feed.

Globe interactions

• Drag horizontally to rotate.

• Drag vertically to tilt where supported.

• Pinch or wheel to zoom.

• Double-tap/double-click to zoom toward a region.

• Reset control returns to the default global camera.

• Tap/click a marker or cluster to open a location preview.

• Zooming progressively changes marker granularity.

Location preview

```
■ LONDON
24 active experiences
183 participants
+4,280 XP earned

[ Explore London ]
```

3. Geographic Hierarchy & Clustering

The globe must use hierarchical aggregation. Never render every experience as an individual marker at world scale. Humanity has already produced enough visual clutter without the app joining in.

Zoom level	Primary representation	Expected behaviour
Global	Countries / major activity clusters	Aggregate activity by geography
Regional	Cities / regions	Show city-level markers and clusters
Local	Districts / nodes / experience clusters	Show increasingly specific activity
Experience	Individual experiences	Show exact experience cards / details

4. Globe Entities

Experiences — Upcoming, active and recently completed real-world experiences.

Nodes — Verified community infrastructure: schools, libraries, community centres, studios, NGOs, universities, businesses, etc.

Communities — Geographic communities with meaningful participation and activity.

Impact — Aggregate contribution signals such as verified contributions, XP and completed experiences.

Location statistics

```
London
4,281 XP
126 experiences
873 participants
312 verified contributions
```

5. Marker System

Markers communicate activity density without turning the globe into a fireworks display. Use marker size, glow, opacity and clustering to represent relative activity.

• Low activity: small, restrained marker.

• Medium activity: larger marker with stronger presence.

• High activity: prominent marker / cluster.

• Very high activity: strongest visual weight, used sparingly.

• Use Experiences.fun visual language: pink #f20789, purple #6805f2, aqua #04adbf, white typography.

• Animations should be subtle and meaningful, not continuous decorative motion.

6. Location Surface

Selecting a city or geographic cluster opens a dedicated location surface. The globe is a navigation mechanism, not a dead-end visualization.

```
LONDON
24 Experiences • 183 Participants • 31 Organisations • 4,281 XP earned

[ Experiences ] [ People ] [ Nodes ]

UPCOMING
Hackney Park Cleanup
```

```
Community Coding Workshop
Photography Walk

ACTIVE COMMUNITY
HydeOut Studios
```

7. Filters & Discovery Controls

• **Primary:** All, Experiences, Communities, Nodes, Impact.

• **Category:** Education, Environmental, Community.

• **Time:** Today, This week, This month.

• **Distance:** Nearby, Anywhere.

• Keep the first release simple. Add complexity only when usage demonstrates demand.

Near You

The page may use a user-selected city/profile location or approximate location when permission is available. Precise location access must not be required for the core globe experience.

```
NEAR YOU
■ London
12 experiences this week
[ Explore ]
```

Trending locations

Trending should be based on activity momentum rather than simply selecting the world's biggest cities. Eventually, use real contribution signals instead of clicks.

```
01 London +42% activity
02 Mogadishu +31% activity
03 Nairobi +24% activity
```

8. Globe + Social Feed Architecture

The Globe and social feed should share the same underlying activity graph. Do not build two disconnected systems that happen to display similar information.

```
ACTIVITY GRAPH
      /   \
   FEED   GLOBE
    WHO    WHERE
     \    /
    EXPERIENCE
```

• Feed answers WHO is doing something.

• Globe answers WHERE it is happening.

• Experience answers HOW to participate.

• The activity graph connects people, places, experiences, accountability and impact.

Activity pulse

A future live layer can surface meaningful activity such as “London +1 experience completed.” Use this sparingly. The product should feel alive, not like a Bloomberg terminal designed by a caffeinated intern.

9. Data Model

```
locations
---------
id
name
type
country_code
latitude
longitude
parent_id
metadata
created_at
```

Supported location types: **country, region, city, district, node, experience**.

Experiences should reference a location_id, allowing hierarchical traversal: **experience** → **location** → **city** → **country**.

10. API Contract

```
GET /api/explore/globe
GET /api/explore/locations/:id
GET /api/explore/locations/:id/experiences
GET /api/explore/locations/:id/activity
GET /api/explore/trending
```

Example globe response

```
{ "locations": [ { "id": "london", "name": "London", "latitude": 51.5072, "longitude": -0.1276,
"experienceCount": 24, "participantCount": 183, "activityScore": 82 } ] }
```

The initial globe endpoint returns aggregate geographic data, not thousands of individual experiences. Fetch detailed data progressively as the user zooms into a region.

11. Frontend Architecture

```
components/explore/
    ExploreGlobe.tsx
    GlobeMarker.tsx
    GlobeCluster.tsx
    LocationPreview.tsx
    LocationStats.tsx
    ExploreFilters.tsx
    TrendingLocations.tsx
    NearbyExperiences.tsx

hooks/
    useExploreGlobe.ts
    useLocations.ts
    useGlobeExperiences.ts
    useLocationActivity.ts
```

Separation of concerns: ExploreGlobe should render geographic data. Business logic, fetching, ranking, aggregation and permissions belong in hooks/services/API layers.

12. Performance & Loading

• Use hierarchical lazy loading: World → Country → City → Local clusters → Individual experiences.

• Do not render all experiences on initial load.

• Use viewport/zoom-aware fetching to control geographic detail.

• Cache aggregate location data aggressively.

• Debounce rapid camera/zoom changes before fetching.

• Cluster markers server-side or client-side depending on scale and renderer performance.

• Use loading, empty and error states for location panels and experience lists.

• Maintain a fast fallback list/grid Explore experience if the 3D renderer is unavailable or too slow.

13. Privacy, Data Quality & Accessibility

• Never expose a user's precise private location as a public marker.

• Experience locations should use intended public venue/node coordinates.

• Use city/area-level approximations where exact coordinates are unnecessary.

• Do not require location permission to use Explore.

• Provide a list-based geographic fallback for users who cannot or do not want to use the 3D globe.

• Ensure markers, controls and previews have accessible labels and adequate contrast.

• Avoid duplicate or stale geographic entities through a clear source of truth and validation pipeline.

• Prevent spam or fake activity from distorting activity density and trending calculations.

14. Visual Direction

• Dark, premium, minimal, spatial and slightly futuristic.

• Black/deep-space background with pink, purple and aqua accents.

• Geographic forms should remain legible and relatively restrained.

• Avoid an overly photorealistic Earth. Prioritise geography, hierarchy and activity.

• Markers should feel native to Experiences.fun, not like generic map pins.

• Use motion to communicate state and interaction, not as decoration.

15. Implementation Phases

Phase	Scope
1 — MVP	Interactive 3D globe; country/city markers; experience clustering; marker → preview; location → experience list; basic filters; nea
2 — Activity	Node markers; community activity; activity density; trending locations; location statistics; richer globe animations.
3 — Impact	Impact visualization; real-time activity pulses; contribution/reputation layers; community health; global challenges; historical activ

16. Acceptance Criteria

• Explore opens with a functional interactive globe.

• Users can rotate and zoom without excessive lag.

• Global view uses aggregated markers/clusters rather than individual experience pins.

• Selecting a marker reveals useful location statistics and an actionable Explore Location path.

• Zooming into an area reveals progressively more granular activity.

• Location pages can show experiences and reuse existing experience card UI.

• Filters affect visible geographic activity.

• Nearby content works without requiring precise location permission.

• Globe and social feed consume compatible activity concepts/data.

• No private user location is exposed.

• There is a non-globe fallback for accessibility/performance.

• Loading, empty and error states are defined.

• Activity density cannot be trivially gamed by low-quality or duplicate events.

17. Agent Build Order

1. Inspect the current Explore screen, navigation, experience card components, data hooks and API conventions.

2. Identify existing experience/location fields and determine the minimum geographic schema needed.

3. Create the location aggregation API and stable response types.

4. Implement ExploreGlobe as a rendering-only component.

5. Implement marker and cluster primitives.

6. Add zoom-aware data fetching and caching.

7. Add LocationPreview and route into a Location Surface.

8. Reuse existing experience cards rather than creating a second card system.

9. Add filters and Nearby/Trending sections.

10. Add fallback list mode, loading states, error states and accessibility labels.

11. Test performance with synthetic high-density geographic data before enabling large-scale production data.

12. Only after MVP stability, add nodes, impact layers and live activity pulses.

18. Long-Term Product Model

The globe should become one of the core interfaces through which Experiences.fun represents the real world:

```
EXPERIENCES.FUN
     ■ THE WORLD
          ■
   PEOPLE / PLACES / EXPERIENCES
          ■
        ACTIVITY
          ■
        IMPACT
          ■
      REPUTATION
```

Final product principle: Explore should not be “another page with cards and a different background.” It should become the geographic operating surface of Experiences.fun, connecting people, places, experiences, activity and eventually measurable real-world impact.